

questions

**Product Name:** BUDGETCHAIN

**2. Product Description: Describe the product in 25 words or less.**

A blockchain-based public budgeting platform that ensures transparent, verifiable, and accountable government fund allocation through immutable records, multi-layer validation, and citizen oversight.

**3. Product Origin: How did you come up with the idea?**

BUDGETCHAIN was conceptualized in response to Senate Bill 1330, which promotes the adoption of blockchain technology to enhance transparency and accountability in government budgeting. The idea was further inspired by real-world governance issues such as the “ghost students” problem, where public funds are allocated to beneficiaries who do not actually exist due to inaccurate or manipulated records.

While blockchain ensures immutability, BUDGETCHAIN explicitly recognizes that it cannot prevent collusion or inaccuracies at the point of data encoding—particularly when encoders come from the same local office. Blockchain only secures data *after* it is recorded; it does not guarantee the truthfulness of the initial input. This limitation, observed alongside delayed disclosures, fragmented budget data, insider collusion risks, and restricted public access, underscored the need for a holistic solution. As a result, BUDGETCHAIN integrates blockchain with multi-layer data validation, cross-office verification (e.g., requiring confirmation from independent provincial or national units), institutional controls, legal compliance mechanisms, and citizen oversight tools that allow anomalies to be flagged publicly—reducing dependence on a single encoder and discouraging coordinated manipulation before data entry.

**4. Problem: What is the problem in the market that your product solves?**

Despite existing transparency portals and audit mechanisms, government budget information remains fragmented, highly technical, and often inaccessible to the public. Data is commonly released in static formats such as PDFs, making verification and analysis difficult.

Additionally, centralized systems remain vulnerable to inaccurate data entry, insider collusion, weak beneficiary verification, and delayed audits, weakening public oversight and increasing the risk of inefficiency, mismanagement, and corruption.

**5. Solution: How does your product solve the problem in Q4 above?**

BUDGETCHAIN leverages blockchain as a secure and immutable record, strengthened by governance and validation mechanisms:

- Privacy-Preserving Multi-Layer Data Validation (Data Privacy Act–Compliant):
  - In line with the Philippine Data Privacy Act of 2012 (RA 10173), particularly the principles of data minimization, proportionality, and purpose limitation, BUDGETCHAIN does not store personal or sensitive information directly on the blockchain. Instead, only anonymized references, aggregated budget figures, and time-stamped cryptographic proofs are recorded on-chain, while identifiable and editable personal data remain off-chain in compliant government databases where correction, updating, or erasure is legally enforceable. This ensures transparency and auditability without violating data subjects’ rights.
- Cross-Institutional Verification Process:
  - To address the risk of pre-encoding manipulation and insider collusion, BUDGETCHAIN assigns verification to independent institutions (e.g., provincial, regional, or national oversight bodies such as COA-linked or treasury-affiliated units), rather than a separate encoder within the same local office. These

verifiers cross-check submissions against independent source documents and standardized budget classifications, ensuring that accuracy is validated beyond the originating office and that manipulated data cannot simply be “re-approved” internally.

- Immutable Blockchain Ledger: Records all stages of the budget lifecycle—allocation, release, disbursement, and expenditure.
- Post-Entry Flagging: Auditors and citizens may flag suspicious records without altering on-chain data.
- Role-Based Access Control & Segregation of Duties: Prevents insider abuse and collusion.
- Automated Audit Trails: Smart contracts log all user actions and trigger audits when anomalies are detected.
- Public Dashboard: A user-friendly interface allowing real-time tracking and verification of government spending.

**6. Customer: Describe the specific customer demographic interested in your product.**

Primary Customers:

- Philippine government institutions, including national agencies, local government units (LGUs), barangays, GOCCs, and state universities responsible for managing public funds.

Ultimate Beneficiaries:

- Filipino citizens, journalists, researchers, and civil society organizations who gain access to clear, accurate, and verifiable public budget information.

**7. Market Size: What is the total size of the market for your product? What source did you use to determine your market size?**

The potential market for BUDGETCHAIN PH includes the entire Philippine public sector, covering national government agencies, local government units (LGUs), barangays, state universities, and government-owned and controlled corporations (GOCCs). Indirectly, the system serves over 110 million Filipino citizens who are stakeholders in public spending. The platform also supports the growing digital governance and public accountability sector, driven by national efforts toward e-government and open data initiatives.

Source: Philippine Statistics Authority (PSA) population data and national digital transformation programs.

**8. Price: What is the price range for your product? How did you determine the price range?**

BUDGETCHAIN is designed as a public service platform, free and accessible to all citizens. Funding is provided by the national government as part of transparency and digital transformation initiatives.

Estimated implementation cost: ₱50–₱200 million, covering software development, blockchain infrastructure, cybersecurity, personnel training, cloud services, and long-term maintenance. Costs are optimized through open-source technologies, phased rollout, and long-term savings from reduced corruption and improved audit efficiency.

**9. Competitors: List your product’s competitors (manufacturer, brand name, model number, and substitutes).**

The DICT blockchain sandbox for the 2026 General Appropriations Act (GAA) is the closest related initiative.

Rather than competing, BUDGETCHAIN is designed to complement and expand this pilot by:

- Extending implementation to barangay-level budgeting
- Providing standardized public dashboards
- Integrating citizen reporting and escalation mechanisms
- Supporting long-term scalability and institutionalization

**10. Advantage: Does the product already exist? If so, how does your product differ from competing products?**

Although budget transparency platforms exist, BUDGETCHAIN offers key advantages:

- Expanded Scope: Designed for scalability from city to national levels, with future extension to barangays once system robustness is established.
- Stronger Data Integrity: Multi-layer validation beyond basic blockchain immutability
- Verified Beneficiary System: Integration with PSA, PhilSys, DepEd, and DSWD databases using unique digital identifiers
- Fraud Prevention: Prevents ghost beneficiaries through traceable, cross-agency verification
- Automated Audit Trails: Smart contracts reduce human error and detect anomalies
- Citizen Empowerment: Formal reporting, flagging, and escalation pathways

**11. Launch: Briefly describe the steps necessary to design, produce, and distribute your product to your first customer.**

Phase 1: Planning and Design

- Define system architecture, permissioned blockchain framework, APIs, and dashboards
- Align with COA, DBM, and DICT standards

Phase 2: City-Level Pilot (Mayor-Led Implementation)

- Train government personnel on encoding, verification, and monitoring
- Test beneficiary validation and audit triggers

Phase 3: Evaluation and Feedback

- Gather feedback from officials, auditors, and citizens
- Refine system features and policies

Phase 4: Gradual Expansion

- Scale to municipalities, cities, provinces, and national agencies

Phase 5: Public Awareness and Accessibility

- Citizen education campaigns
- Mobile-friendly dashboards and low-bandwidth access
- Barangay kiosks and simplified offline reports

Phase 6: Continuous Monitoring and Security

- Regular cybersecurity audits and penetration testing
- System updates, stress testing, and policy sandboxing

Attachments: Are any attachments provided with this submission form? Please check.

- ☐ Drawings
- ☐ Sketches
- ☐ Photographs
- ☒ Any other information





BUDGETCHAIN

GAA 2026 INFRASTRUCTURE

OPERATIONS

CHAIN LEDGER

AGENCIES

REGISTRY

AUDIT LOGS

OVERSIGHT

DOST\_CORE\_HUB

SEARCH CHAIN (TX HASH, PROJECT ID, OR DOCUMENT ROOT)...

BAGONG PILIPINAS DIGITAL TRANSPARENCY PORTAL

AU AUDITOR\_PRIME COA\_AUTH\_01

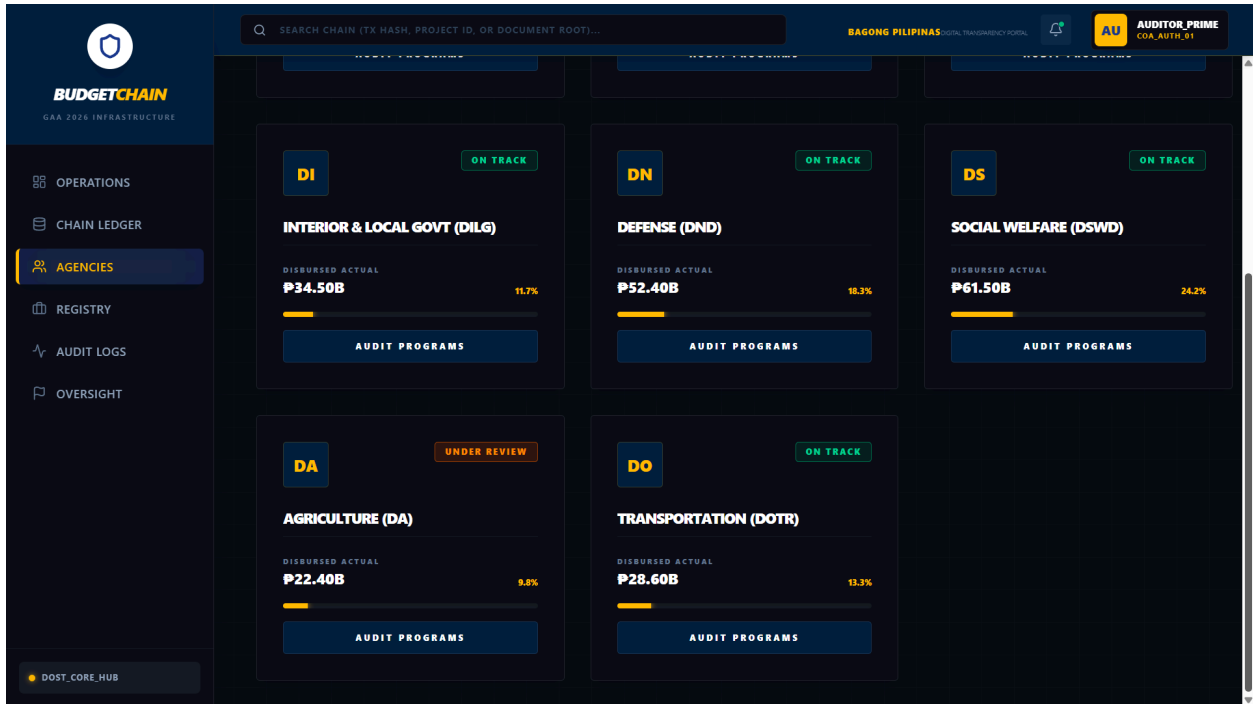
LEDGER EXPLORER

IMMUTABLE ALLOTMENT REGISTRY (FY 2026 CYCLE)

FILTER

EXPORT DATASET

| ANCHOR HASH   | INSTRUMENT | AGENCY PROFILE | CERTIFIED VALUE | VALIDATION | TIMESTAMP      |
|---------------|------------|----------------|-----------------|------------|----------------|
| 0xaf4e...b219 | GAA_2026   | DOTR           | P94.50B         | ANCHORED   | JAN 2026 CYCLE |
| 0x7d1a...cc04 | GAA_2026   | DSWD           | P126.80B        | ANCHORED   | JAN 2026 CYCLE |
| 0x2b1c...ff92 | GAA_2026   | DEPED          | P22.40B         | ANCHORED   | JAN 2026 CYCLE |
| 0x8f2d...92a1 | GAA_2026   | DPWH           | P18.20B         | ANCHORED   | JAN 2026 CYCLE |
| 0x1c4b...3b11 | GAA_2026   | DOH            | P35.80B         | ANCHORED   | JAN 2026 CYCLE |





**BUDGETCHAIN**  
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BAGONG PILIPINAS - DIGITAL TRANSPARENCY PORTAL

  **AUDITOR\_PRIME**  
COA\_AUTH\_01



### ADMINISTRATIVE LOGS

DL\_MANIFEST



**GAA 2026 ANCHORED**

JAN 01, 2026

FINAL SIGNED BUDGET P6.750T SUCCESSFULLY ANCHORED TO GENESIS BLOCK FOR FY 2026

 AUTHORIZED SIGNER: **DBM\_SYSTEM\_NODE**



**Q1 NCA ISSUED**

5 HOURS AGO

P145B NCA RELEASED FOR SOCIAL PROTECTION AND HEALTH PROGRAMS

 AUTHORIZED SIGNER: **ADMIN\_DBM\_PRIME**



**BUDGETCHAIN**  
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DOST\_CORE\_HUB

SEARCH CHAIN (TX HASH, PROJECT ID, OR DOCUMENT ROOT)...

BAGONG PILIPINAS - DIGITAL TRANSPARENCY PORTAL

  **AUDITOR\_PRIME**  
COA\_AUTH\_01

## ACCOUNTABILITY NODE

CITIZEN-LED ACCOUNTABILITY MATRIX. REPORTS ARE SIGNED AND ANCHORED PERMANENTLY TO THE FISCAL GENESIS BLOCK FOR COA REVIEW.

INITIALIZE DISCREPANCY INQUIRY



**OPEN RESOLUTIONS**



ZERO ACTIVE ESCALATIONS DETECTED IN BLOCK EPOCH 1042.



**CONSENSUS HUB STATUS**



**NCR NATIONAL NODE (DBM)**

0c0a...ff2



**REGION VII REGIONAL NODE**

0c0b...d01



**REGION XI REGIONAL NODE**

Syncing Blocks...

FORCE HUB SYNCHRONIZATION

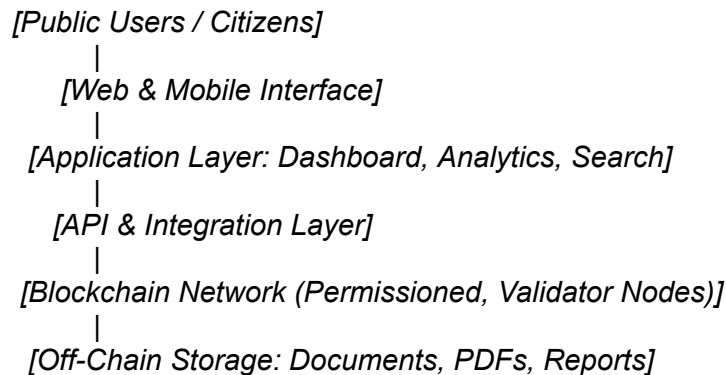
attachments



## System Architecture Diagram

A layered architecture showing how data flows from government agencies to the blockchain network and to the public dashboard.

### Text-based Diagram:



### Explanation:

- **Public Users:** Citizens view budget data in real-time.
- **Web & Mobile Interface:** Dashboard, search, and visualization tools.
- **Application Layer:** Handles queries, analytics, and reporting.
- **API Layer:** Integrates with existing government financial systems.
- **Blockchain Network:** Stores immutable budget transaction records.
- **Off-Chain Storage:** Stores large supporting documents; only cryptographic hashes are recorded on-chain.

## Blockchain Workflow

### Step-by-Step Process:

1. Budget allocation, fund release, or disbursement occurs.
2. Authorized government official submits transaction data.
3. Validator nodes (barangay, municipal, or national-level agencies) verify the transaction.
4. Transaction is hashed and linked to previous blocks.
5. Consensus is reached and the block is added to the blockchain.
6. Transaction is automatically displayed on the public dashboard.

### Simplified Flowchart:

Submit → Validate → Hash → Approve → Record → Display

## Public Interface Layout

**Home Screen:** National budget overview, charts, live blockchain activity.

**Departments Section:** List of all agencies with budget allocations and expenditures.

**Programs & Projects Section:** Detailed project budgets, progress, and status.

**Transaction Explorer:** Searchable blockchain records with timestamps and supporting documentation.

**Regional View:** Interactive map showing allocations by barangay, city, or province.

**Feedback/Reporting:** Citizens can submit reports or concerns linked to specific transactions.

## Data Governance Framework

### Components:

1. **Data Ownership:** Government owns the data; blockchain ensures verifiable records.
2. **Access Control:**

| User Type               | Access Level                   |
|-------------------------|--------------------------------|
| Citizens/Public         | View-only                      |
| Researchers/Journalists | View + Download Reports        |
| Government Staff        | Submit & Validate Transactions |
| Auditors                | Full Audit Access              |

3.  
**Data Privacy:** Sensitive personal information is stored off-chain; only cryptographic hashes are on-chain.
4. **Data Integrity & Security:** Immutable blockchain, permissioned nodes, cryptographic verification.
5. **Compliance:** Aligned with **Senate Bill 1330**, **Data Privacy Act**, and government cybersecurity policies.

## File Upload Process for Government Agencies

1. Secure Login
  - Each agency (national, provincial, municipal, or barangay) has **authorized accounts**.
  - Login uses **two-factor authentication (2FA)** to ensure that only verified personnel can submit files.
2. Access Submission Portal
  - Once logged in, the user navigates to the “**Submit Budget/Project Data**” section.
  - This portal is integrated with the **application layer**, which automatically prepares data for blockchain processing.
3. Uploading Files
  - Agencies can upload supporting documents such as:
    - Budget allocation spreadsheets
    - Fund release vouchers

- Project plans or receipts
- Reports on beneficiaries (e.g., student lists, community project reports)
- Supported formats: **PDF, XLS/XLSX, CSV, and images (JPEG/PNG)**
- Maximum file size can be configured for efficiency.

#### 4. Automatic File Hashing

- When a file is uploaded:
  - The system generates a **cryptographic hash** of the file.
  - This hash is **recorded on the blockchain**, ensuring the file cannot be altered without detection.
  - The file itself is stored **off-chain** on a secure government server or cloud storage.

#### 5. Data Verification

- Validator nodes (other government offices) review the submission for accuracy.
- Only verified files are **linked to a blockchain transaction**.
- Any changes require a new submission with a new hash; the original record remains immutable.

#### 6. Confirmation and Public Display

- Once approved, the transaction appears on the **public dashboard**.
- Citizens can see:
  - Transaction ID (hash)
  - Timestamp
  - File reference
- They **cannot alter or delete the file**, ensuring transparency.

#### 7. Optional Automation

- **Smart contracts** can automate certain submissions:
  - Automatic record of fund releases or allocations
  - Trigger public display once validation is complete

- This reduces manual errors and speeds up transparency.